UNIVERSITY OF ZAGREB



Robotic Sensing, Perception, and Actuation

Stereo Visual Odometry

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January 13, 2026

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1 Abstract

This report presents a comprehensive analysis of a visual odometry (VO) system implementing monocular-to-stereo pose estimation using ORB (Oriented FAST and Rotated BRIEF) feature detection on the KITTI benchmark dataset. The system employs three distinct approaches: baseline monocular VO, constrained optimization with depth uncertainty filtering, and bundle adjustment refinement. Evaluation metrics (APE - Absolute Pose Error and RPE - Relative Pose Error) demonstrate the effectiveness of feature filtering and depth-aware constraints in improving trajectory accuracy across 11 KITTI sequences.

2 Introduction

2.1 Problem Statement

Accurate ego-motion estimation is essential for autonomous navigation and robotics. Visual odometry (VO) estimates the camera trajectory from sequential images and provides a low-cost alternative to LiDAR-based localization.

2.2 Motivation

- **Feature-based VO:** robust to illumination changes compared to purely photometric methods.
- **ORB features:** fast extraction and matching, enabling real-time performance.
- **Stereo + KITTI:** stereo geometry provides metric depth; KITTI offers a standard benchmark for evaluation.

2.3 Contributions

1. A baseline stereo VO pipeline using 3D–2D PnP with RANSAC.
2. A constrained pipeline with spatial keypoint selection, rectified-stereo filtering, and depth-uncertainty rejection.
3. Quantitative evaluation on KITTI sequences, including runtime and pose error analysis.

3 Methodology

This project implements a real-time stereo visual odometry (VO) pipeline on the KITTI Odometry benchmark. We estimate the camera motion between consecutive stereo frames using a standard 3D–2D formulation: (1) detect and match features, (2) triangulate 3D landmarks from stereo, and (3) estimate the relative pose via PnP with RANSAC. Two variants are evaluated: a baseline pipeline and a constrained pipeline that adds feature filtering and motion priors to improve robustness and trajectory stability.

3.1 Dataset and Notation

For each sequence, rectified grayscale stereo pairs are processed sequentially. We denote the left and right images at time k as (I_k^L, I_k^R). The KITTI calibration file provides the projection matrices $\mathbf{P}^L$ and $\mathbf{P}^R$ as well as the intrinsics $\mathbf{K}^L$ and $\mathbf{K}^R$. The goal is to estimate the camera pose over time as a sequence of rigid transforms in SE(3).

3.2 Pipeline Overview

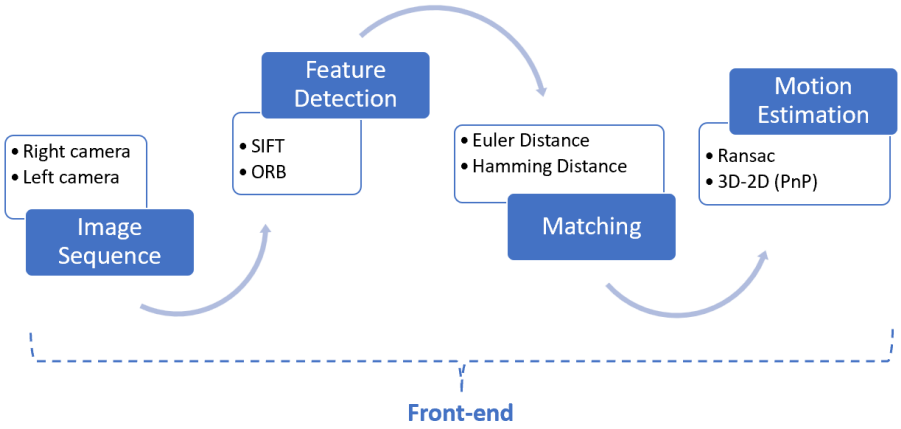


Figure 1: Visual Odometry Front-end

3.3 Feature Detection and Descriptor Choice

We evaluated SIFT and ORB for feature detection/description. SIFT typically provides more distinctive descriptors and robust matches, but its computational cost is significantly higher. ORB provides binary descriptors with fast Hamming-distance matching and is therefore better suited for real-time VO. Based on the observed runtime difference on KITTI sequences, ORB was selected for all subsequent experiments.

Sequence	SIFT (iters/s)	ORB (iters/s)
0	5.93	35.59
4	3.63	37.35
7	3.84	36.12

Table 1: Runtime comparison between SIFT and ORB (iterations per second).

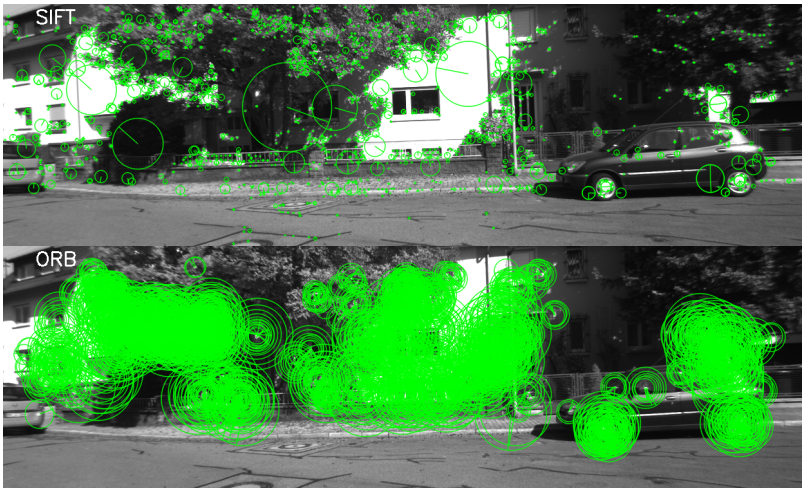


Figure 2: Qualitative comparison of detected keypoints: SIFT (top) vs ORB (bottom) on KITTI sequence 00.

3.4 Feature Matching

We use brute-force matching with Hamming distance for ORB descriptors. Two match sets are computed at each step:

- **Temporal matches** $\mathcal{M}_{\text{temp}}$: between I^L_{k-1} and I^L_k.
- **Stereo matches** $\mathcal{M}_{\text{st}}$: between I^L_{k-1} and I^R_{k-1}.

Only features that are valid in both sets are retained, ensuring each track has both a temporal correspondence and a stereo correspondence (required for triangulation + 3D–2D PnP).

3.4.1 Baseline Matching

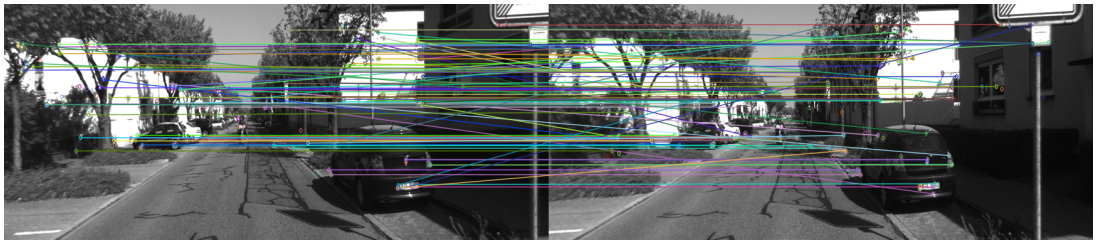
The baseline uses brute-force descriptor matching (optionally with cross-checking) and a distance threshold. This approach is simple but can produce geometrically inconsistent stereo matches in challenging regions, which leads to invalid triangulation.

3.4.2 Rectified Stereo Matching with Geometric Constraints

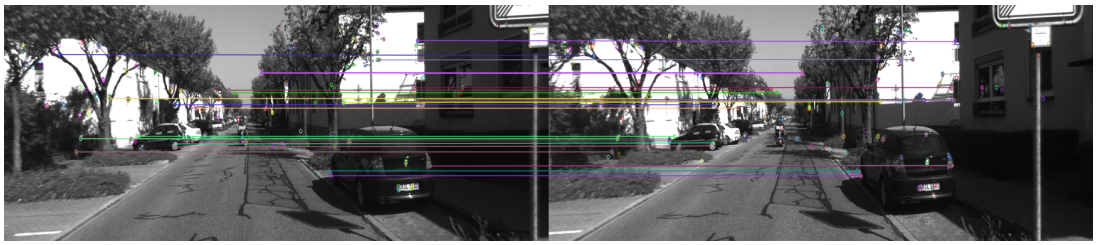
The constrained pipeline makes stereo matching more selective by enforcing rectified-stereo geometry:

- **Epipolar constraint:** $|v^L - v^R| \leq \tau_v$ (small vertical pixel tolerance).
- **Disparity bounds:** $d = u^L - u^R \in [d_{\min}, d_{\max}]$.
- **Descriptor filtering:** Hamming threshold + ratio test (Lowe-style).
- **Optional cross-check:** keep only mutual best matches.

This reduces the total number of matches but substantially decreases incorrect correspondences and improves the stability of triangulation and PnP.



(a) Brute-force stereo matching



(b) Rectified stereo matching

Figure 3: Stereo matching comparison: baseline brute-force matching vs. rectified stereo matching with geometric filtering.

3.5 Stereo Triangulation

Given stereo correspondences $(\mathbf{x}^L_{k-1}, \mathbf{x}^R_{k-1})$ from time $k-1$, 3D landmarks are reconstructed via triangulation using $\mathbf{P}^L$ and $\mathbf{P}^R$. The resulting 3D points $\mathbf{X}_{k-1}$ are expressed in the camera frame at time $k-1$ and paired with their corresponding 2D observations $\mathbf{x}^L_k$ in the current left image for 3D–2D pose estimation.

3.6 Pose Estimation via PnP with RANSAC

The relative motion from $k-1$ to k is estimated by solving a Perspective-n-Point problem with RANSAC:

$$\min_{\mathbf{R},\mathbf{t}} \sum_i \left\| \mathbf{x}_{k,i}^L - \pi\big(\mathbf{K}^L(\mathbf{R}\mathbf{X}_{k-1,i} + \mathbf{t})\big) \right\|^2,$$

where $\pi(\cdot)$ denotes the projection onto the image plane. RANSAC reduces the influence of mismatches by estimating the pose from inliers only.

The estimated incremental motion is integrated into a global pose estimate by compounding transforms over time. If too few valid correspondences remain, the update for that frame is skipped and the previous pose is reused to avoid injecting large outliers into the trajectory.

3.7 Robustness Enhancements in the Constrained Pipeline

3.7.1 Spatial Keypoint Selection (Grid-Based Clustering)

To avoid keypoint over-concentration in highly textured areas, we enforce a more uniform spatial distribution. The image is divided into fixed-size grid cells; within each cell, keypoints are sorted by response and only the strongest few are retained. This reduces redundant, correlated measurements and improves numerical conditioning for triangulation and PnP.

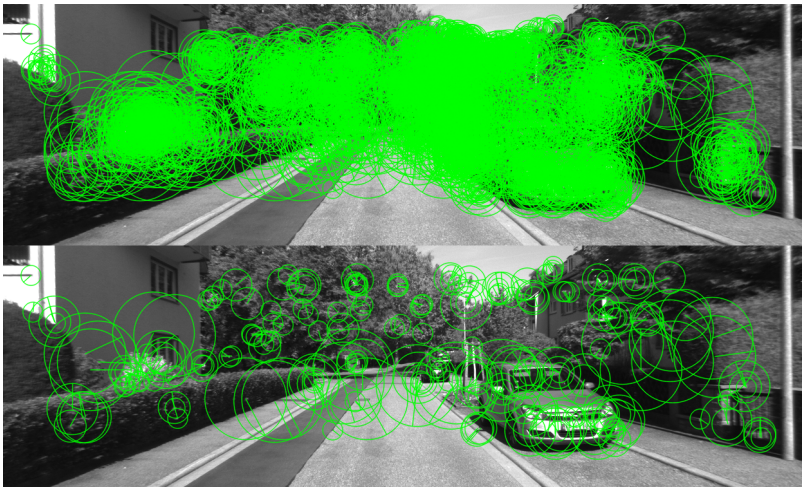


Figure 4: Spatial keypoint selection. Original ORB keypoints (top) vs. grid-filtered keypoints (bottom).

3.7.2 Depth-Uncertainty Filtering

Triangulation becomes unreliable for far landmarks with very small disparity. Assuming a disparity uncertainty of approximately one pixel, first-order propagation yields:

$$\sigma_z \approx \frac{z^2}{f\,b},$$

where z is depth, f the focal length, and b the stereo baseline. We retain points that are in front of the camera and satisfy a relative uncertainty bound:

$$\frac{\sigma_z}{z} < 0.5.$$

Optionally, a minimum depth threshold (e.g., $z_{\min} = 3b$) is applied to reject near-degenerate geometry.

3.7.3 Planar Motion Prior and Outlier Rejection

Vehicle motion in KITTI is approximately planar; therefore, roll and pitch in the estimated rotation are often dominated by noise and weak geometric conditioning. After PnP, the rotation is first projected to the closest valid rotation using SVD (orthonormalization). A yaw-only rotation $\mathbf{R}_{\text{planar}}$ is then constructed by extracting the yaw angle from the orthonormalized rotation:

$$\theta = \arctan 2(R_{0,2}, R_{0,0}), \qquad \mathbf{R}_{\text{planar}} = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}.$$

A soft constraint is applied by blending the planar rotation with the estimated rotation and projecting the result back onto SO(3) using SVD. In addition, heuristic bounds reject frames with unrealistically large translation magnitude or extreme pitch/roll, which are strong indicators of PnP failure.

3.8 Optional Post-Processing: Bundle Adjustment

As an additional refinement (when enabled), bundle adjustment (BA) is applied as a post-processing step to optimize camera poses (and optionally landmarks) over a local window or selected keyframes. This reduces accumulated drift by jointly minimizing reprojection error across multiple frames, at the cost of increased runtime.

3.9 Algorithm Summary

Algorithm 1 Stereo Visual Odometry (Baseline / Constrained)

Load calibration $(\mathbf{K}^L, \mathbf{K}^R, \mathbf{P}^L, \mathbf{P}^R)$
Initialize pose $\mathbf{T}_0 \leftarrow \mathbf{I}$
Detect ORB features and descriptors in (I_0^L, I_0^R)
for $k = 1$ to $N - 1$ **do**
    Load (I_k^L, I_k^R) and detect ORB features in I_k^L
    if constrained **then**
        Apply grid-based keypoint selection in I_k^L (and I_{k-1}^L, I_{k-1}^R as needed)
    end if
    Temporal match: $I_{k-1}^L \leftrightarrow I_k^L$
    Stereo match: $I_{k-1}^L \leftrightarrow I_{k-1}^R$
    if constrained **then**
        Enforce rectified stereo constraints and descriptor ratio test
    end if
    Triangulate 3D points $\mathbf{X}_{k-1}$ from stereo matches
    if constrained **then**
        Filter $\mathbf{X}_{k-1}$ using depth uncertainty and validity checks
    end if
    if $|\mathbf{X}_{k-1}|$ is too small **then**
        Skip update; keep previous pose
        continue
    end if
    Estimate $(\mathbf{R}, \mathbf{t})$ via PnP-RANSAC using $\mathbf{X}_{k-1}$ and I_k^L observations
    if constrained **then**
        Apply planar-motion prior and outlier rejection checks
    end if
    Integrate pose: $\mathbf{T}_k \leftarrow \mathbf{T}_{k-1}\,\Delta\mathbf{T}(\mathbf{R}, \mathbf{t})$
    Store $\mathbf{T}_k$
end for

3.10 Key Parameters Used

Parameter	Baseline	Constrained	Notes
ORB `nfeatures`	3000	3000	Target number of features
ORB `scaleFactor`	1.6	1.2	Pyramid scale step
ORB `nlevels`	8	8	Pyramid levels
Grid size (px)	–	50×30	Spatial keypoint selection
Max keypoints / cell	–	7	Keep strongest by response
Stereo vertical tol. τ_v (px)	–	2.0	Rectified epipolar tolerance
Ratio test	–	0.8	Lowe-style descriptor ratio
Depth rule	–	$\sigma_z/z < 0.5$	Uncertainty-based filtering
Min depth	–	$z_{\min} = 3b$	Optional (baseline b)

Table 2: Representative parameter settings used in the baseline and constrained pipelines.

4 Results and Discussion

This section evaluates the proposed visual odometry system. We first compare the baseline and constrained pipelines on KITTI Sequence 0 to highlight the impact of geometric and motion constraints. We then evaluate the constrained pipeline on three sequences with different motion profiles using trajectory, orientation, and error metrics.

4.1 Baseline vs Constrained on Sequence 0

KITTI Sequence 0 contains frequent turns and viewpoint changes, making it well suited to reveal drift. Figure 5 shows the estimated x–z trajectories. The baseline exhibits accumulated drift and local distortions, particularly in curved segments and intersections. The constrained pipeline produces a smoother trajectory that better preserves loops and sharp turns, indicating improved geometric consistency from the added stereo filtering, depth rejection, and planar motion prior.

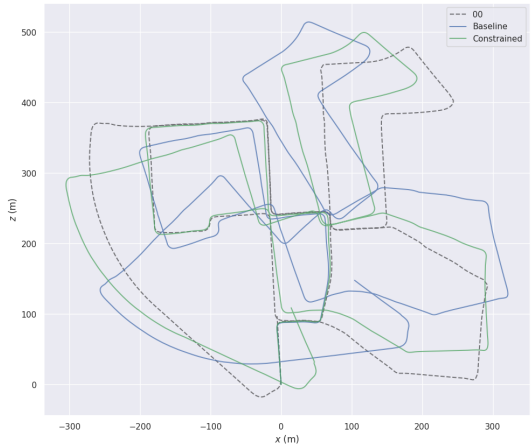


Figure 5: x–z trajectory comparison between the baseline (blue) and constrained (green) pipelines on KITTI Sequence 0

4.2 Evaluation Sequences

The constrained pipeline is evaluated on three KITTI sequences:

- **Seq. 0**: urban driving with frequent turns and loops.
- **Seq. 1**: highway driving with long straight segments and sparse static features.

- **Seq. 5**: mixed urban motion with moderate turns.

Together, they cover both turning and mostly straight motion under different levels of scene structure.

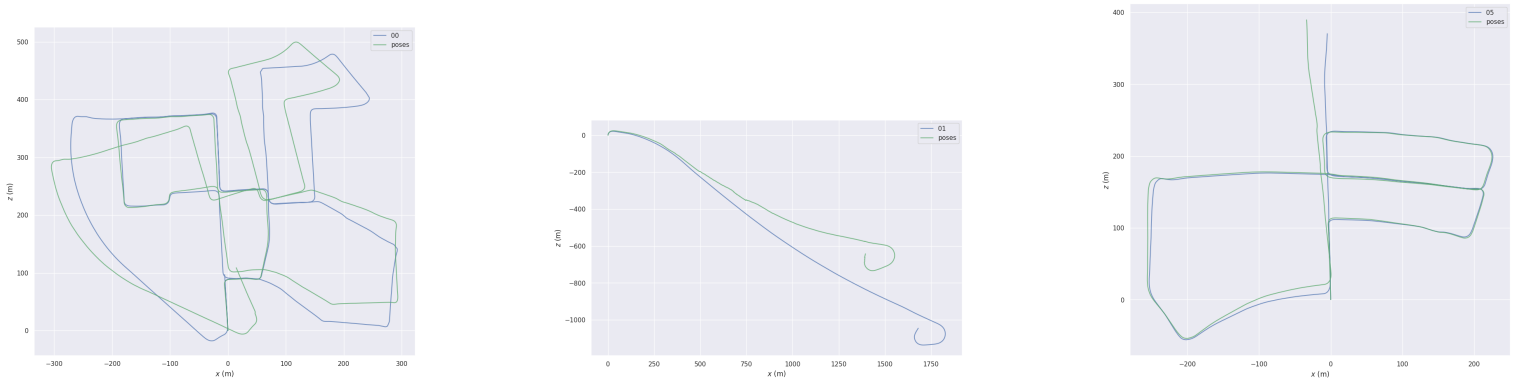
All experiments use a fixed parameter set:

$$\begin{aligned}&\text{ORB: } n_{\text{features}} = 3000,\ \text{scaleFactor} = 1.2,\ n_{\text{levels}} = 8; \quad \text{Grid: } (50, 30),\ K = 7;\\ &z_{\min} = 3b; \quad \|\mathbf{t}\|_{\max} = 15\,\text{m},\ \text{yaw}_{\max} = 30^\circ; \quad \Delta v_{\max} = 2,\ d_{\max} = 64;\\ &\text{cross-check} = \text{true},\ \text{Hamming}_{\max} = 50,\ \text{ratio} = 0.8.\end{aligned}$$

4.3 Trajectory Analysis

Figure 6 compares the x–z trajectories of the constrained pipeline against ground truth for Sequences 0, 1, and 5.

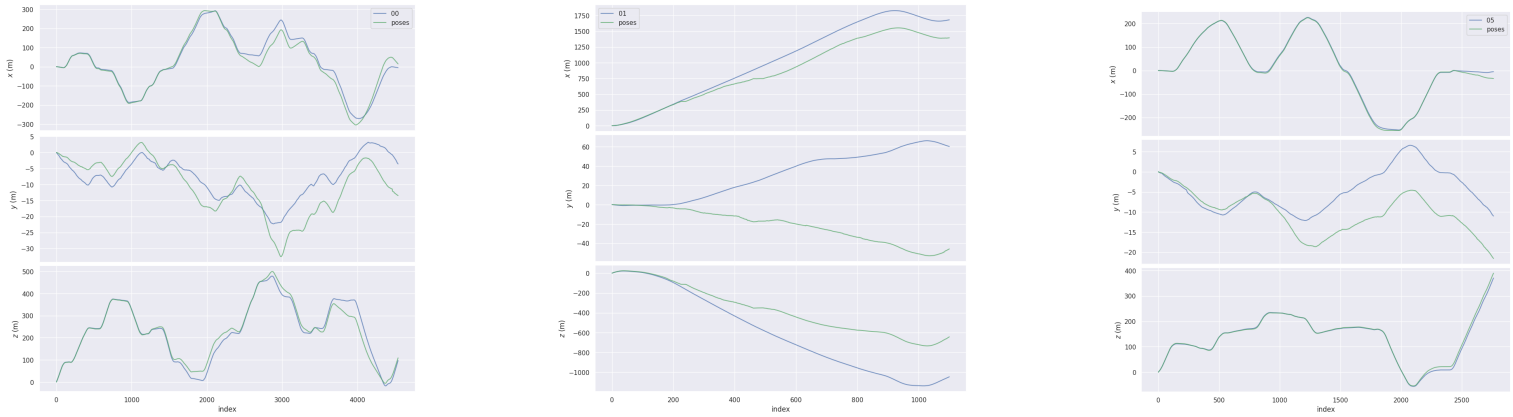
In **Seq. 0**, the main loops and turns are recovered, but drift accumulates after long travel and repeated turns. In **Seq. 1**, the estimate diverges strongly due to weak geometric constraints in long straight highway motion and additional outliers from moving vehicles. In **Seq. 5**, the trajectory closely matches ground truth, indicating sufficient static structure and parallax for stable stereo pose estimation.



(a) (b) (c)

Figure 6: x–z trajectories for Sequences 0, 1, and 5 using the constrained pipeline

Figure 7 shows the estimated x, y, and z components for Sequences 0, 1, and 5. The largest deviations occur in y, which is expected for ground vehicles since motion is mostly planar and vertical parallax is limited, making y more sensitive to triangulation and pose noise.



(a) (b) (c)

Figure 7: x, y, and z position estimates for Sequences 0 (a), 1(b), and 5 (c) using the constrained pipeline.

4.4 Orientation Analysis

Orientation consistency can be assessed by examining how well the estimated trajectory preserves the direction of straight segments and the alignment of turns. Across all evaluated sequences, yaw tracking was found to be highly accurate, which is reflected in the strong agreement between the estimated and ground-truth trajectories in the horizontal plane. Figure 8 illustrates yaw estimate for Sequence 5.

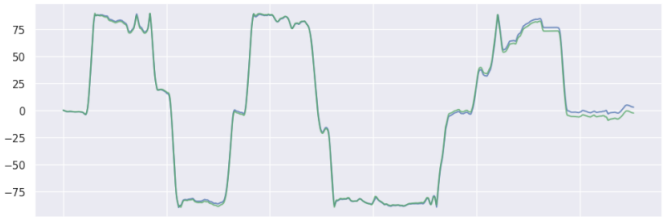


Figure 8: Yaw angle for Sequence 5 (green) comapred to reference value (blue).

Similar behavior was observed in Sequences 0 and 1, where the vehicle heading remained stable over long straight segments and through curved sections. This strong yaw consistency explains the good alignment of the estimated x–z trajectories with the ground truth across all sequences.

4.5 Quantitative Error Metrics

Table 3 reports the Absolute Pose Error (APE) and Relative Pose Error (RPE) for the constrained pipeline. APE is measured over the full trajectory, while RPE is computed over 100 m segments.

Sequence	Length (km)	RMSE APE (m)	Max APE (m)	RMSE RPE (m)	Max RPE (m)
0	3.7	39.01	87.59	1.71	4.29
1	2.45	346.55	531.25	61.03	151.40
5	2.2	11.81	36.18	1.74	3.83

Table 3: APE and RPE for the constrained pipeline on selected KITTI sequences

4.6 Discussion

The constrained pipeline improves robustness over the baseline, but performance still depends on scene structure. Seq. 5 achieves low drift thanks to strong static features and parallax, while Seq. 0 shows moderate drift from accumulated yaw error over long trajectories. Seq. 1 degrades severely in highway motion due to weak structure and dynamic objects that harm triangulation. Overall, trajectories are smoother than the baseline, but long-term consistency likely requires additional cues such as loop closure or sensor fusion.

5 Conclusion

We developed and evaluated a real-time stereo visual odometry pipeline on the KITTI dataset using a 3D–2D PnP formulation. After comparing SIFT and ORB, ORB was selected for its significantly higher runtime performance. A constrained variant was then introduced using spatial keypoint selection, rectified-stereo geometric filtering, uncertainty-aware depth filtering, and a planar motion prior, which reduced mismatches and improved trajectory stability, especially in urban sequences with frequent turns, while maintaining real-time operation. Quantitative results show low relative pose error in feature-rich scenes but degraded performance on highway segments due to limited parallax and unreliable depth, reflecting known limitations of front-end-only VO. Future work will integrate bundle adjustment and loop closure to further reduce drift and move toward full SLAM.